C++ Mock 1 Batch_FEB_2026

Total points 30/62

The respondent's email (**vaibhavkharche4328@gmail.com**) was recorded on submission of this form.

What is the correct syntax to declare a pure virtual function in a C++ class? *1/1

- (selected, correct) A) virtual void func() = 0;
- B) virtual void func();
- C) void func() = 0;
- D) virtual void func()

Which of the following statements is correct about a friend function in C++? *1/1

- A) It can be a member of another class.
- (selected, correct) B) It can access private and protected members of a class.
- C) It is a member of the class in which it is declared.
- D) It cannot be used to access private members of a class.

What does the following C++ code print? * 1/1

```
#include<iostream>

using namespace std;


class MyClass {

public:

   int x;

   MyClass() : x(10) {}

   void setX(int val) { x = val; }

   void printX() { cout << x << endl; }

};


int main() {

   MyClass obj;

   obj.setX(20);

   obj.printX();

}
```

- A) 10
- B) 20 ✓
- C) Error
- D) 0

Which of the following is true about operator overloading in C++? * 0/1

- A) Operator overloading allows the modification of the behavior of operators in C++.
- B) You cannot overload operators for primitive data types.
- C) Operator overloading can change the precedence of operators. (selected, incorrect)
- D) Operators can only be overloaded for classes with one data type

Correct answer

- A) Operator overloading allows the modification of the behavior of operators in C++.

Which of the following statement will be correct if the function has three arguments passed to it? * 0/1

- The trailing argument will be the default argument.
- The first argument will be the default argument.
- The middle argument will be the default argument.
- All the argument will be the default argument. (selected, incorrect)

Correct answer

- The trailing argument will be the default argument.

Which one of the following is the correct way to declare a pure virtual function? *0/1

- virtual void Display(void){0};
- virtual void Display = 0;
- virtual void Display(void) = 0;
- void Display(void) = 0;

Correct answer

- virtual void Display(void) = 0;

Which of the following is **incorrect** regarding the this pointer? * 1/1

- a) The this pointer can be used to differentiate between local variables and class members when they have the same name.
- b) The this pointer refers to the address of the current object.
- c) The this pointer is available only in non-static member functions.
- d) The this pointer is passed explicitly to each member function by the compiler.

Which of the following statement is not true about C++? * 1/1

- A class cannot have the data members as pointer.
- Dynamic objects in c++ can be created only with the help of new operator.
- Using abstract class we can achieve runtime polymorphism.
- Static function can be overrided.

Which of the following is the default access modifier for members of a class in C++? *1/1

- (selected) A) private ✓
- B) public
- C) protected
- D) None of the above

Which one of the following statements correctly refers to the Delete and Delete[] in C++ programming language? *1/1

- The "delete" is used for deleting the standard objects, while on the other hand, the "Delete[]" is used to delete the pointer objects
- The "Delete" is a type of keyword, whereas the "Delete[]" is a type of identifier
- (selected) The "Delete" is used for deleting a single standard object, whereas the "Delete[]" is used for deleting an array of the multiple objects ✓
- Delete is syntactically correct although, if the Delete[] is used, it will obtain an error

Which of the following functions must use the reference in the argument list to avoid chain of calls? *1/1

- (selected) Copy constructor ✓
- Friend Function
- Operator Function
- Default Constructor

Which of the following statements is correct about the friend function in C++ programming language? *1/1

- A friend function can access the private members of a class
- A friend function is able to access protected members of a class
- A friend function is able to access the public members of a class
- All of the above

What is the output of the following C++ code snippet? * 0/1

int a[5] = {1, 2, 3, 4, 5};

cout << a[2] << a[3] << a[4];

- a) 12345
- b) 2345
- c) 234
- d) 345

Correct answer

- b) 2345

What will be the output of the following C++ code? * 1/1

```
#include<iostream>
using namespace std;

class Test {
public:
    int x;
    Test(int val) { x = val; }
};

int main() {
    Test obj(10);
    cout << obj.x << endl;
}
```

- A) 0
- B) 10
- C) Error
- D) Undefined behavior

In C++, when a class member is declared private, which of the following is true? *0/1

- A) The private member can be accessed outside the class by using an object of the class.
- B) The private member can be accessed only by member functions within the class.
- C) The private member can be accessed only within the class constructor.
- D) The private member can be accessed by any function in the program.

Correct answer

- B) The private member can be accessed only by member functions within the class.

What is the size of an object of class Cdate when the class contains three integers and one character array of size 10? *1/1

```
#include<iostream>

using namespace std;

class Cdate {

   int dd, mm, yy;

   char name[10];

public:

   void accept() {}

   void display() {}

};

int main() {

   Cdate d1;

   cout << sizeof(d1) << endl;

}
```

- A) 10
- B) 16
- C) 20
- D) 24

What will be the output of the following code? * 1/1

```
#include<iostream>
using namespace std;

class MyClass {
public:
  MyClass() {
    cout << "Constructor Called" << endl;
  }
  ~MyClass() {
    cout << "Destructor Called" << endl;
  }
};

int main() {
  MyClass obj;
}
```

- (selected) A) Constructor Called Destructor Called ✓
- B) Destructor Called Constructor Called
- C) Error
- D) No output

What is the output of the Program? * 0/1

```
class Base
{
   int x, y;
   public:
   int z;
   public:
   Base()
   {
      x = y = z = 0;
   }
   void Display(void)
   {
      cout<< x << " " << y << " " << z << endl;
   }
};
class Derived : public Base
{
   int x, y;
   public:
   Derived(int xx = 65, int yy = 66)
   {
      y = xx;
      x = yy;
   }
   void Display(void)
   {
      cout<< x << " " << y << " "<<z;

   }
};
int main()
{
   Derived objD;
   objD.Display();
   return 0;
}
```

- 000
- 66 65 0

66 65 garbage

none of the above.

Correct answer

66 65 0

Assume that Honda is an instance of the Car class, and that Car class has a member function named run. Which of the following is a correct call to the run function? * 1/1

Honda->run;

Honda.run();

run()

Honda()

If a class has no constructor, which of the following will happen? * 1/1

A) The class will throw an error during object creation.

B) A default constructor will be automatically provided.

C) A default constructor will not be created automatically, and the object cannot be created.

D) The class will not be able to access member variables.

Which of the following statements is true about default arguments in C++? *0/1

- a) Default arguments must be provided from right to left in the function declaration.
- b) Once a default argument is used in a function, all subsequent arguments must have default values.
- c) Default arguments can be overridden when calling the function.
- d) All of the above.

Correct answer

- d) All of the above.

Which of the following statements about constructors and destructors in C++ is false? *1/1

- A) Constructors are used to initialize objects.
- B) Destructors are used to destroy objects and release resources.
- C) A constructor can be called explicitly by the user.
- D) A class can have multiple constructors with different parameter lists (overloaded constructors).

In C++, what will be the output if the following code is executed? * 1/1

```
#include<iostream>
using namespace std;

class Complex {
private:
    int real, img;

public:
    Complex(int r, int i) : real(r), img(i) {}
    Complex() {
        real = 0;
        img = 0;
    }
    void display() {
        cout << real << "+" << img << "i" << endl;
    }
};

int main() {
    Complex c1(5, 3), c2;
    c1.display();
    c2.display();
}
```

- A) 5+3i 0+0i

- B) 5+3i Error
- C) 0+0i 5+3i
- D) Error

What is the output of the following C++ code? * 1/1

```
#include<iostream>
using namespace std;

class Rectangle {
  int length, width;
public:
  void setData(int l, int w) {
    length = l;
    width = w;
  }
  int area() { return length * width; }
};

int main() {
  Rectangle rect;
  rect.setData(10, 5);
  cout << "Area: " << rect.area() << endl;
}
```

- (●) A) Area: 50 ✓
- ( ) B) Area: 15
- ( ) C) Area: 10
- ( ) D) Error

Which of the following C++ features provides the ability to use the same function name with different arguments? *1/1

- A) Inheritance
- B) Encapsulation
- C) Polymorphism
- D) Abstraction

What is the output of the following code? * 1/1

```
#include<iostream>
using namespace std;

class Base {
    int x;
public:
    void setX(int val) { x = val; }
    int getX() { return x; }
};

class Derived : public Base {
    int y;
public:
    void setY(int val) { y = val; }
    int getY() { return y; }
};

int main() {
    Derived obj;
    obj.setX(10);
    obj.setY(20);
    cout << obj.getX() << " " << obj.getY() << endl;
}
```

- (selected) A) 10 20 ✓

B) 0 0

C) Error

D) 10 0

Which of the following is the correct declaration for a constant pointer to an integer in C++? * 0/1

A) int* const ptr;

B) const int* ptr;

C) int const* ptr;

D) const ptr int*;

Correct answer

A) int* const ptr;

Which of the following statement is correct? * 0/1

Overloaded functions can accept same number of arguments.

Overloaded functions always return value of same data type.

Overloaded functions can accept only same number and same type of arguments.

Overloaded functions can accept only different number and different type of arguments.

Correct answer

Overloaded functions can accept same number of arguments.

PRN *

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Which of the following type of class allows only one object of it to be created? * 1/1

- ○ Virtual class
- ○ Abstract class
- ◉ Singleton class ✓
- ○ Friend class

Which one of the following cannot be used with the virtual keyword? * 0/1

- ◉ Destructor ✗
- ○ Member function
- ○ Constructor
- ○ None of the above

Correct answer

- ◉ Constructor

Which of the following statements about constant objects is **incorrect**? * 0/1

- a) A constant object can invoke both constant and non-constant member functions.
- b) Constant objects can invoke only constant member functions.
- c) The data members of a constant object cannot be modified.
- d) Constant objects are read-only objects.

Correct answer

- a) A constant object can invoke both constant and non-constant member functions.

In C++, which of the following statements about destructors is true? * 1/1

- A) A destructor is automatically called when an object goes out of scope.
- B) A destructor cannot take arguments.
- C) A destructor can be overloaded.
- D) A destructor is used to initialize object properties.

Which of the following C++ features allows a class to inherit members and behaviors from another class? * 1/1

- A) Encapsulation
- B) Polymorphism
- C) Inheritance
- D) Abstraction

In C++, what is the correct way to declare a constant array of integers? * 0/1

- a) const int arr[5] = {1, 2, 3, 4, 5};
- b) int const arr[5] = {1, 2, 3, 4, 5};
- c) Both a and b
- d) int arr[5] const = {1, 2, 3, 4, 5};

Correct answer

- c) Both a and b

What will be the output of the following C++ code? * 0/1

```
#include <iostream>
using namespace std;
int main ()
 {
 char str1[10] = "Hello";
 char str2[10] = "World";
 char str3[10];
 int len ;
 strcpy( str3, str1);
 strcat( str1, str2);
 len = strlen(str1);
 cout<<len<<endl;
 return 0;
 }
```

- (selected) 5 ✗
- 12
- 10
- 11

Correct answer

- (selected) 10

What happens when you try to call a non-constant member function on a constant object? * 0/1

- a) It will compile successfully, but the function cannot modify the object.
- b) It will result in a compilation error.
- c) The function will be executed, but it will be restricted to read-only operations.
- d) It will cause undefined behavior.

Correct answer

- b) It will result in a compilation error.

What is the purpose of static variables in a function in C++? * 1/1

- a) They retain their values across multiple calls to the function.
- b) They only exist during the execution of the function.
- c) They cannot be initialized within the function.
- d) They are used for memory allocation.

Find the output of the following program. * 0/1

```
int main()
{
    char ch[] = "c++ programs";
    int i = sizeof(ch);
    cout << i << endl;
}
```

- ○ 13
- ◉ 12 ✗
- ○ 1
- ○ 4

Correct answer

- ◉ 13

Name * …/1

Vaibhav Ashok Kharche ✗

Which of the following definition best describes the concept of polymorphism? * 1/1

- ○ It is the ability to process the many messages and data in one way
- ○ It is the ability to process the undefined messages or data in at least one way
- ◉ It is the ability to process the message or data in more than one form ✓
- ○ It is the ability to process the message or data in only one form

What is the output of the Program? * 0/1

```
#include <iostream>
using namespace std;
class Demo
{
   public:
   Demo(int xx)
   {
      cout<< xx;
   }
   ~Demo()
   {
      cout<< "Final";
   }
};
int main()
{
  Demo *ptr = new Demo('B');
   return 0;
}
```

- (selected) Compile time error ✗
- B
- garbage
- 66

Correct answer

- (selected) 66

class Complex { int real, img;public: Complex(int r, int i) : real(r), img(i) {} void display() const { cout << real << "+" << img << "i" << endl; } void setReal(int r) { real = r; }};int main() { const Complex c(1, 2); c.display(); c.setReal(3); // Error?}
What will happen when you attempt to run this code? *0/1

- a) The code will run without any error.
- b) The code will compile successfully but result in undefined behavior.
- c) The code will result in a compilation error.
- d) The code will run but setReal will not modify the object.

Correct answer

- c) The code will result in a compilation error.

What will happen if you try to use the this pointer in a static member function? *1/1

- a) It will compile successfully.
- b) It will result in a compilation error.
- c) The code will run with undefined behavior.
- d) The static member function will automatically receive the this pointer.

What does the following C++ code print? * 0/1

```
#include<iostream>
using namespace std;

int main() {
   int a = 5;
   int b = ++a;
   cout << a << " " << b << endl;
}
```

- A) 5 5
- B) 6 5
- C) 6 6
- D) 5 6

Correct answer

- C) 6 6

NAME *

Vaibhav Ashok Kharche

Which of the following is used for implementing the late binding? * 0/1

- Operator Functions
- Virtual Functions
- new Operator
- Static Functions

Correct answer

- Virtual Functions

Which of the following is not allowed in C++? * 1/1

- A) Declaring multiple constructors in a class
- B) Declaring a function inside a class
- C) Declaring a function outside a class
- D) Declaring a variable of type function

Which of the following can be considered as the correct syntax for declaring an array of pointers of integers that has a size of 10 in C++? *0/1

- int *arr = new int*[10]
- int *arr = new int[10];
- int arr = new int[10];
- int **arr = new int*[10];

Correct answer

- int **arr = new int*[10];

PRN *

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What will be the output of the following C++ program? * 1/1

```
void display(char c = '*', int count = 30);

int main() {

  int count = 80;

  display('#');

  display('$', count);

  display();

  return 0;

}

void display(char c, int count) {

  for(int i = 1; i <= count; ++i) {

    cout << c;

  }

  cout << endl;

}
```

- (selected) a) The program will print '#' followed by 80 '$' characters and then a line of '' characters.
- b) The program will print '#' followed by 30 '$' characters and then a line of '' characters.
- c) The program will print '*' followed by 30 '#' characters and then a line of '$' characters.
- d) The program will cause an error since default arguments are used in an incorrect manner.

Which of the following is **not** a valid declaration for a constant member function? *0/1

- (selected, incorrect) a) int getData() const;
- b) void display() const { cout << "Display"; }
- c) void setData(int x) const { data = x; }
- d) void modify() const;

Correct answer

- c) void setData(int x) const { data = x; }

Consider the following C++ code: * 0/1

int arr[5] = {1, 2};

cout << arr[2] << arr[3] << arr[4];

What is the output?

- a) 120
- b) 2 3 4
- (selected, incorrect) c) Garbage values
- d) Compilation error

Correct answer

- a) 120

What is the output of the Program? * 1/1

```
#include <iostream>
using namespace std;
class Demo
{
   int x, y;
   public:
   void SetValue(int &a, int &b)
   {
      a = 100;
      x = a;
      y = b;
      Display();
   }
   void Display()
   {
      cout<< x << " " << y;
   }
};
int main()
{
   int x = 10;
   Demo d;
   d.SetValue(x, x);
   return 0;
}
```

- ( ) The program will print the output 100 10.
- (•) The program will print the output 100 100. ✓
- ( ) The program will print the output 100 garbage.
- ( ) It will result in a compile time error.

In C++, which of the following is true about the sizeof operator when used on an array? *0/1

- a) It gives the number of elements in the array.
- b) It gives the size in bytes of the array.
- c) It is only valid for static arrays, not dynamic arrays.
- d) Both b and c.

Correct answer

- d) Both b and c.

In C++, what happens when you try to access an array element outside its bounds? *0/1

- a) Compiler throws an error.
- b) Compiler performs bound checking at runtime.
- c) Undefined behavior occurs.
- d) The program crashes immediately.

Correct answer

- c) Undefined behavior occurs.

What is the correct syntax to pass an array to a function in C++? * 0/1

- a) func(arr[]);
- b) func(int arr[]);
- c) func(arr[5]);
- d) func(int* arr);

Correct answer

- d) func(int* arr);

Consider the following code snippet:
class Employee { int id; char name[30];public: void getdata() { cout << "Enter Id: "; cin >> id; cout << "Enter Name: "; cin >> name; } void putdata() const { cout << id << " " << name << endl; }};int main() { const Employee e; e.getdata(); // Error? e.putdata();}What will happen when you run the code? *1/1

- a) The code will compile and run successfully.
- b) The code will compile but give an error during the getdata function call.
- c) The code will compile and give an error during the putdata function call.
- d) The code will not compile because constant objects cannot call any member functions

What is the difference between a static data member and a non-static data member in a C++ class? *0/1

- a) A static data member is shared by all objects of the class, while a non-static data member is unique to each object.
- b) A static data member can be accessed without creating an object, but a non-static data member cannot.
- c) A non-static data member can be initialized only inside the constructor.
- d) Both a and b are correct

Correct answer

- d) Both a and b are correct

What is the output of the following code? *0/1

```
#include <iostream>  using namespace std;  class Test {  public:      int x;
   Test(int val) { x = val; }      void modify() const { x = 10; }  };  int main() {
   const Test t(5);      t.modify();      cout << t.x;      return 0;  }
```

- a) Compilation error due to modification of constant object
- b) 5
- c) 10
- d) Undefined behavior

Correct answer

- a) Compilation error due to modification of constant object

What is the primary difference between constant objects and constant member functions in C++? *1/1

- a) Constant objects can call both constant and non-constant member functions.
- b) Constant member functions cannot modify the object's data members, while constant objects cannot invoke non-constant functions.
- c) Constant objects can modify the data members but cannot call non-constant member functions.
- d) There is no difference between constant objects and constant member functions.

Which of the following type of data member can be shared by all instances of its class? *0/1

- Public
- Protected
- Static
- Friend

Correct answer

- Static

What will happen when you try to modify a constant argument inside a function? *0/1

```
void test(const int x) {    x = 10; // Error?}
```

- a) The code will compile successfully and modify x. (selected, incorrect)
- b) The code will result in a compile-time error due to the attempt to modify a constant argument.
- c) The code will run with undefined behavior.
- d) The code will give a warning but still run successfully.

Correct answer

- b) The code will result in a compile-time error due to the attempt to modify a constant argument.

Which one of the following given statements is not true about the references in C++? *0/1

- Array of reference cannot be created.
- A reference cannot refer to a constant value
- A reference cannot be NULL (selected, incorrect)
- Once a reference is created, it cannot be later made to reference another object; it cannot be reset

Correct answer

- A reference cannot refer to a constant value